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# Contrasting roles for IKK regulated inflammatory signalling pathways for development and maintenance of type 1 and adaptive $\gamma\delta$ T cells

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## eLife Assessment

This study reports **important** findings regarding the role of the NF- $\kappa$ B signaling pathway in the development and long-term survival of gamma delta T cells. The authors report disparate roles of IKK-dependent NF- $\kappa$ B activation in the development and long-term survival of gamma delta T cell subsets. The approach and methodology employed is **convincing**. This work will be of great interest to immunologists interested in innate-like T cell biology and in T cell development.

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## Abstract

The inhibitor of kappa B kinase complex (IKK) is a critical regulator of cell death and inflammatory signaling in multiple cell types. Phosphorylation of I $\kappa$ B proteins by IKK results in their degradation and consequent activation of NF- $\kappa$ B transcription factors. RIPK1, a critical cell death regulator, is also a direct target of IKK kinase activity, thereby repressing its cell death activity. In  $\alpha\beta$  T cells, the RIPK1 kinase activity of IKK is critical for normal thymic development while mature  $\alpha\beta$  T cells require IKK for both activation of NF- $\kappa$ B dependent survival programmes, and repression of RIPK1.  $\gamma\delta$  T cells play a unique and versatile role in host immunity with specific effector functions that enables them to act as early responders in immune defence. The role of IKK regulated pathways in their development and survival is not known. Here, we use mouse genetics to dissect the function of IKK and downstream pathways in the normal homeostasis of  $\gamma\delta$  T cells. We find that IKK expression is critical to establish a replete  $\gamma\delta$  T cell compartment, but that requires vary between different subsets. Type 1  $\gamma\delta$  T cells require IKK dependent NF- $\kappa$ B activation for their generation, while IKK is redundant for development of adaptive  $\gamma\delta$  T cells. Instead, IKK dependent NF- $\kappa$ B activation is required for their longterm survival. We also find evidence that IKK repression of RIPK1 is required for survival of peripheral but not thymic  $\gamma\delta$  T cells. Ablation of CASPASE8 did not rescue  $\gamma\delta$  T cells in the absence of IKK but rather revealed a potent sensitivity of all  $\gamma\delta$  subsets to necroptosis, that was rescued by kinase dead RIPK1. Overall, we reveal critical requirements for IKK regulated inflammatory pathways by  $\gamma\delta$  T cells that contrast with those of  $\alpha\beta$  T cells, and between different subsets, highlighting the complexity of the regulation of these pathways in the adaptive immune system.

## Introduction

The NF- $\kappa$ B family of transcription factors play critical roles in controlling development and function of many cell types (Bonizzi and Karin, 2004 [DOI](#)). Canonical NF- $\kappa$ B signalling is mediated by hetero or homodimers of p50, RELA and cREL family members that are sequestered in the cytoplasm by inhibitory proteins, the Inhibitors of kappa B (I $\kappa$ B) family and the related protein

NF- $\kappa$ B1. The key regulator of NF- $\kappa$ B dimer release is the inhibitor of kappa-B kinase (IKK) complex, a trimeric complex of two kinases, IKK1 (IKK $\alpha$ ) and IKK2 (IKK $\beta$ ), and a third regulatory component, NEMO (IKK $\gamma$ ). IKK phosphorylates I $\kappa$ B proteins, targeting them for degradation by the proteasome and releasing NF- $\kappa$ B dimers to enter the nucleus.

In  $\alpha\beta$  T cells, activation of NF- $\kappa$ B is a well-recognised critical early event during T cell receptor antigen recognition (Gerondakis and Siebenlist, 2010). In the absence of REL subunits, or upstream NF- $\kappa$ B activators, such as TAK1 or IKK complex, T cells fail to blast transform or enter cell cycle (Webb et al., 2019; Xing et al., 2016). Consequently, mice with T cell-specific ablation of RELA and/or cREL have peripheral naive T cells, but lack effector or memory phenotype T cells (Webb et al., 2019; Zheng et al., 2003). As well as regulating activation, the NF- $\kappa$ B signaling pathway also has important functions in the development and survival of  $\alpha\beta$  T cells. While activation of NF- $\kappa$ B by TCR appears redundant for normal selection of thymocytes during thymic development (Schmidt-Suppran et al., 2004; Webb et al., 2019), signals from TNF and other TNF receptor superfamily members (TNFRSF) are important for survival and differentiation of post selection single positive thymocytes. NF- $\kappa$ B is required for re-expression of *Il7r* following thymic development, that is necessary for long term IL-7 dependent survival of peripheral naive T cells (Miller et al., 2014; Silva et al., 2014). Additionally, long-term survival of fully mature naive CD4<sup>+</sup> T cells is dependent upon tonic NF- $\kappa$ B since CD4<sup>CreERT</sup> induced deletion of REL subunits results in a substantial loss of T cells (Carty et al., 2023).

In addition to activating NF- $\kappa$ B dependent transcriptional activity to regulate T cell survival, TNF-induced NF- $\kappa$ B signalling pathways also control cell death through the activity of IKK. In addition to its function as an inhibitor of kappa-B kinase, the IKK complex also directly controls cell survival independent of NF- $\kappa$ B activation. While thymic development is largely normal in the absence of REL subunits (Webb et al., 2019), IKK deficiency in  $\alpha\beta$  T cells results in a profound block in thymic development that is arrested at the SP stage (Schmidt-Suppran et al., 2003) due to TNF-induced apoptosis (Webb et al., 2016). Ligation of TNFR1 causes recruitment of TRADD, TRAF2, and the serine/threonine kinase RIPK1. The ubiquitin ligases TRAF2, cellular inhibitor of apoptosis proteins (cIAPs) and the linear ubiquitin chain assembly complex (LUBAC), add ubiquitin chain modifications to themselves and RIPK1, creating a scaffold that allows recruitment and activation of the TAB/TAK and IKK complexes that in turn activate NF- $\kappa$ B. This is termed complex I (Annibaldi and Meier, 2018; Vandenabeele et al., 2010). A failure to maintain the stability of this complex results in the formation of cell death inducing complexes composed of TRADD, FADD, CASPASE8 and RIPK1 that induce apoptosis, a function dependent upon RIPK1 kinase activity (Annibaldi and Meier, 2018; Dondelinger et al., 2016; Ting and Bertrand, 2016). Phosphorylation of RIPK1 by IKK blocks RIPK1 kinase activity and therefore its capacity to induce apoptosis (Dondelinger et al., 2015). In thymocytes, it is this function of IKK, and not NF- $\kappa$ B activation, that is critical for their survival and onward development and accounts for the phenotype observed in IKK deficiency (Blanchett et al., 2022; Webb et al., 2019). Survival of mature T cells depends upon both the IKK capacity to repress RIPK1 and its function to activate NF- $\kappa$ B (Carty et al., 2023). RIPK1-dependent cell death of thymocytes and mature naive T cells is mediated by apoptosis, rather than necroptosis, since cell death is CASPASE8 dependent (Carty et al., 2023). Thymocytes are not susceptible to necroptosis as they lack expression of MLKL (Webb et al., 2019), a key effector molecule for mediating necroptotic cell death. Similarly, resting peripheral  $\alpha\beta$  T cells are resistant to necroptotic cell death, since *Casp8* deletion has little impact on peripheral  $\alpha\beta$  T cell compartments (Ch'en et al., 2008; Ch'en et al., 2011). In contrast, activated T cells are acutely susceptible to necroptosis and readily undergo necroptotic cell death in the absence of *Casp8* expression during LCMV infection or following activation in vitro (Ch'en et al., 2008; Ch'en et al., 2011).

$\gamma\delta$  T cells are a distinct subset of T lymphocytes that play a unique and versatile role in host immunity. While conventional  $\alpha\beta$  T cells rely on MHC-restricted antigen presentation,  $\gamma\delta$  T cells recognise stress-induced ligands, phosphoantigens, and non-peptidic molecules directly (Deseke

and Prinz, 2020 [\[10\]](#)). Distinct subsets of innate-like  $\gamma\delta$  T cells develop with specific effector functions, such as type 1 and type 17  $\gamma\delta$  T cells (Munoz-Ruiz et al., 2017 [\[11\]](#); Vantourout and Hayday, 2013 [\[12\]](#)).

Weak TCR signalling during thymic development is associated with generation of type 17  $\gamma\delta$  T cells, that are  $CD44^{hi}$  and lack CD27 expression, and develop as a finite wave in a very specific window of embryogenesis (Munoz-Ruiz et al., 2017 [\[11\]](#)). In contrast, type 1  $\gamma\delta$  T cell development is associated with strong TCR signalling and cells assume a  $CD122^{hi}CD27^{+}$  phenotype during thymic development. The remainder of  $\gamma\delta$  T cells in lymphoid tissues are undifferentiated and exhibit a more naive  $CD44^{lo}CD27^{+}$  phenotype, and are termed by some as naive or adaptive  $\gamma\delta$  T cells.

The pre-differentiated type 1 and type 17 states enable them to act as early responders in immune defence. As such, while it seems likely that inflammatory NF- $\kappa$ B and cell death signalling pathways would be important regulators of  $\gamma\delta$  T cell development and homeostasis, as is the case for  $\alpha\beta$  T cells, very little is currently known. Evidence that NF- $\kappa$ B signalling could be important comes from the observation that the TNFRSF member CD27 is an important regulator of type 1  $\gamma\delta$  T cell development (Ribot et al., 2009 [\[13\]](#)) and is also an activator of NF- $\kappa$ B signalling required for survival of  $\alpha\beta$  T cells (Silva et al., 2014 [\[14\]](#)). However, whether and how inflammatory signalling pathways mediated by IKK and NF- $\kappa$ B for the normal homeostasis of  $\gamma\delta$  T cells remains unknown.

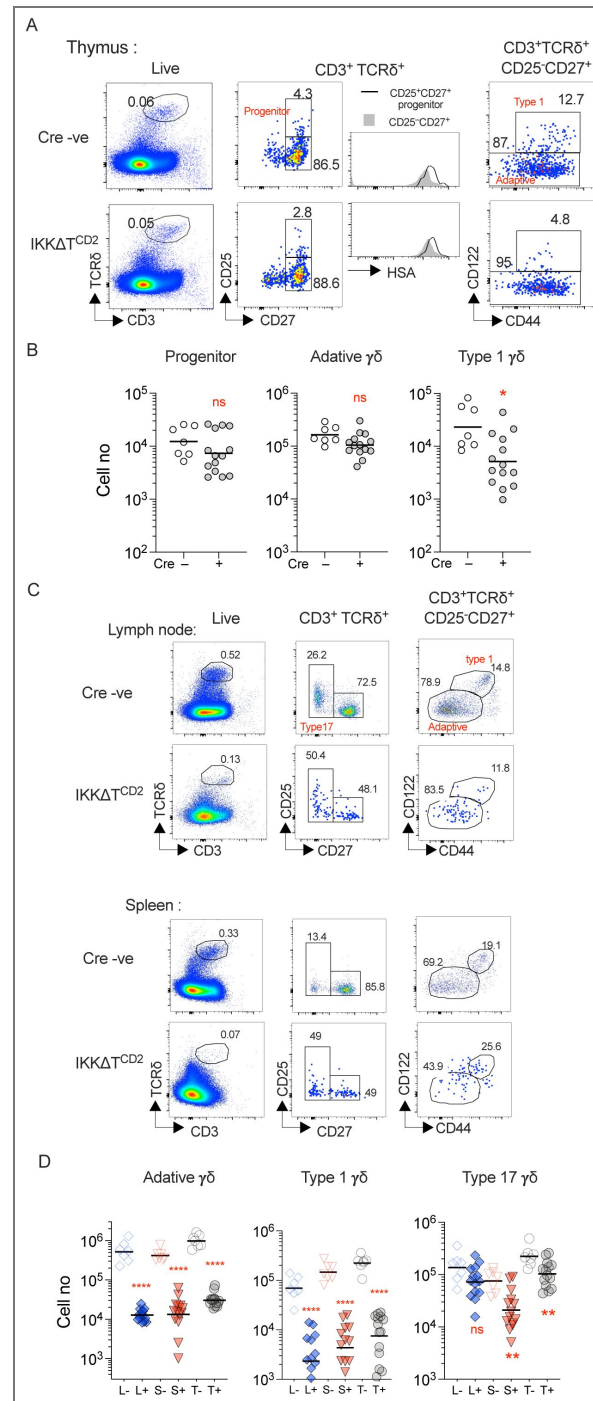
In the present study, we used mouse genetics to investigate the role of IKK and NF- $\kappa$ B survival and cell death pathways for homeostasis of  $\gamma\delta$  T cells. We found evidence that thymic development of type 1  $\gamma\delta$  T cells was dependent on these pathways, but redundant for adaptive  $\gamma\delta$  T cells. However, long-term survival of adaptive  $\gamma\delta$  T cells was dependent upon both NF- $\kappa$ B and, in part, RIPK1-dependent survival pathways and both populations were highly susceptible to necroptotic cell death.

## Results

### IKK signalling is required for generation of type 1 $\gamma\delta$ T cells and maintenance of adaptive $\gamma\delta$ T cells

We first asked whether IKK signalling was required for  $\gamma\delta$  T cell specification and development of naive/adaptive (adaptive hereon) and type 1  $\gamma\delta$  T cells in the thymus of adult mice lacking expression of IKK proteins in the T cell compartment. Type 17  $\gamma\delta$  T cells develop as a finite wave in a very specific window of embryogenesis (Munoz-Ruiz et al., 2017 [\[11\]](#)). Since our analysis was focused on adult genetic mutants we only assessed type 17  $\gamma\delta$  T cells in secondary lymphoid organs. We generated mice with conditional genes encoding IKK1 and IKK2 protein (*Chuk<sup>flox</sup>* and *Ikkb<sup>flox</sup>* respectively) and an iCre transgenic construct controlled by huCD2 expression elements (IKK $\Delta T^{CD2}$  mice hereon). huCD2iCre is expressed in CLP in bone marrow, and Cre-mediated gene deletion is evident almost all of the earliest thymic progenitors that enter the thymus (Siegemund et al., 2015 [\[15\]](#)), so it is ideal to target gene deletion prior to  $\gamma\delta$  T cell specification that occurs during DN2 (Fiala et al., 2020 [\[16\]](#)). In confirmation, we analysed huCD2<sup>iCre</sup> mediated Rosa26RmTom Cre reporter expression in adaptive, type 1 and type 17  $\gamma\delta$  T cell subsets and found ubiquitous expression of reporter in all subsets (Fig. S1). Analysing thymi from IKK $\Delta T^{CD2}$  mice revealed normal representation and numbers of both  $CD25^{+}CD27^{+}TCR\delta^{+}$  progenitor cells and total  $CD25^{-}CD27^{+}TCR\delta^{+}$  T cells, suggesting that IKK signalling was not required for either specification of the  $\gamma\delta$  T cell lineage from uncommitted DN precursors or subsequent thymic development of  $\gamma\delta$  T cells. HSA expression was high in both populations in IKK $\Delta T^{CD2}$  mice, confirming their immature status (Fig. 1A [\[17\]](#)). Type 1  $\gamma\delta$  T cells can be identified by their expression of CD122. Analysing subsets of  $CD25^{-}CD27^{+}TCR\delta^{+}$  T cells revealed that  $CD122^{-}$  adaptive  $\gamma\delta$  T cells were present in normal numbers while numbers of  $CD122^{+}$  type 1  $\gamma\delta$  T cells were significantly and substantially reduced in the absence of IKK activity (Fig. 1B [\[18\]](#)).

The thymic phenotype of IKK $\Delta T^{CD2}$  mice suggested that IKK signalling is required for the development of type 1  $\gamma\delta$  T cells. Analysing numbers of type 1  $\gamma\delta$  T cells recovered from lymph node and spleen confirmed this view since IKK $\Delta T^{CD2}$  mice were almost completely devoid of this



**Figure 1. Development of type 1 and persistence of adaptive and type 17  $\gamma\delta$  T cells depends on IKK expression.**

Thymi, lymph nodes and spleen from IKKΔT<sup>CD2</sup> mice (n=14) and Cre -ve littermates (n=7) were enumerated and analysed by flow. (A) Representative flow plots are of thymocytes with the indicated gates, showing gates used to identify progenitor, adaptive and type 1 subsets in thymically (See Materials and Methods for details). (B) Scatter plots are of total cell the thymus of IKKΔT<sup>CD2</sup> mice or Cre -ve littermates. Horizontal bars are mean. (C) Representative flow plots illustrate the gating strategy to identify adaptive, type 1 and type 17 subsets in lymph nodes and spleen (see Materials and Methods for details). (D) Scatter plots are of total cell numbers recovered from lymph nodes (L), spleen (S) and both spleen and lymph nodes combined (T) of the indicated subset from IKKΔT<sup>CD2</sup> mice (+) or Cre -ve littermates (-). Horizontal bars are mean. Fractional representation of different  $\gamma\delta$  subsets in different organs is shown in Figure S2A. Data are pooled from multiple batches of mice analysed. Horizontal lines indicate mean. \* p<0.05, \*\* p<0.01, \*\*\* p<0.0001, Mann Whitney test.

subset (Fig. 1C-D). Strikingly, adaptive  $\gamma\delta$  T cells were also largely absent from secondary lymphoid organs. Only a small population of CD27<sup>+</sup> type 1  $\gamma\delta$  T cells were detectable in periphery of IKK $\Delta$ T<sup>CD2</sup> mice, but these were also significantly reduced in number compared to Cre<sup>-ve</sup> littermates (Fig. 1D). Together, these data suggest that IKK signalling is required for development of type 1  $\gamma\delta$  T cells and for the long term survival of adaptive  $\gamma\delta$  T cells.

## Additional *Casp8* deficiency rescues generation type 1 $\gamma\delta$ development in the thymus but not maintenance of peripheral $\gamma\delta$ T cells in the absence of IKK

In  $\alpha\beta$  T cells, IKK signalling is required to repress acute induction of *Casp8*-dependent cell death in thymocytes and peripheral T cells. In thymocytes, this survival function is entirely independent of NF- $\kappa$ B, while in mature peripheral T cells, IKK also triggers NF- $\kappa$ B transcriptional activity that contributes to their survival (Carty et al., 2023). To test which functions of IKK are required by different  $\gamma\delta$  T cell populations, we first analysed IKK $\Delta$ T<sup>CD2</sup> mice with additional deletion of *Casp8* (Casp8.IKK $\Delta$ T<sup>CD2</sup> mice). In the thymus, numbers of progenitor  $\gamma\delta$  T cells in Casp8.IKK $\Delta$ T<sup>CD2</sup> mice were similar to those of Cre<sup>-ve</sup> littermates (Fig. 2A), as observed in IKK $\Delta$ T<sup>CD2</sup> mice. In contrast, numbers of adaptive  $\gamma\delta$  T cells in Casp8.IKK $\Delta$ T<sup>CD2</sup> mice exhibited a modest but statistically significant reduction, while numbers of type 1  $\gamma\delta$  T cells appeared to be rescued to levels similar to Cre<sup>-ve</sup> controls, suggesting that development of this subset was *Casp8* dependent in the absence of IKK expression. However, this rescue appeared limited in nature, since analysing  $\gamma\delta$  T cells in the periphery of Casp8.IKK $\Delta$ T<sup>CD2</sup> mice failed to reveal any detectable rescue of cell numbers of any  $\gamma\delta$  T cell subsets in the lymph node and spleen (Fig. 2B-C).

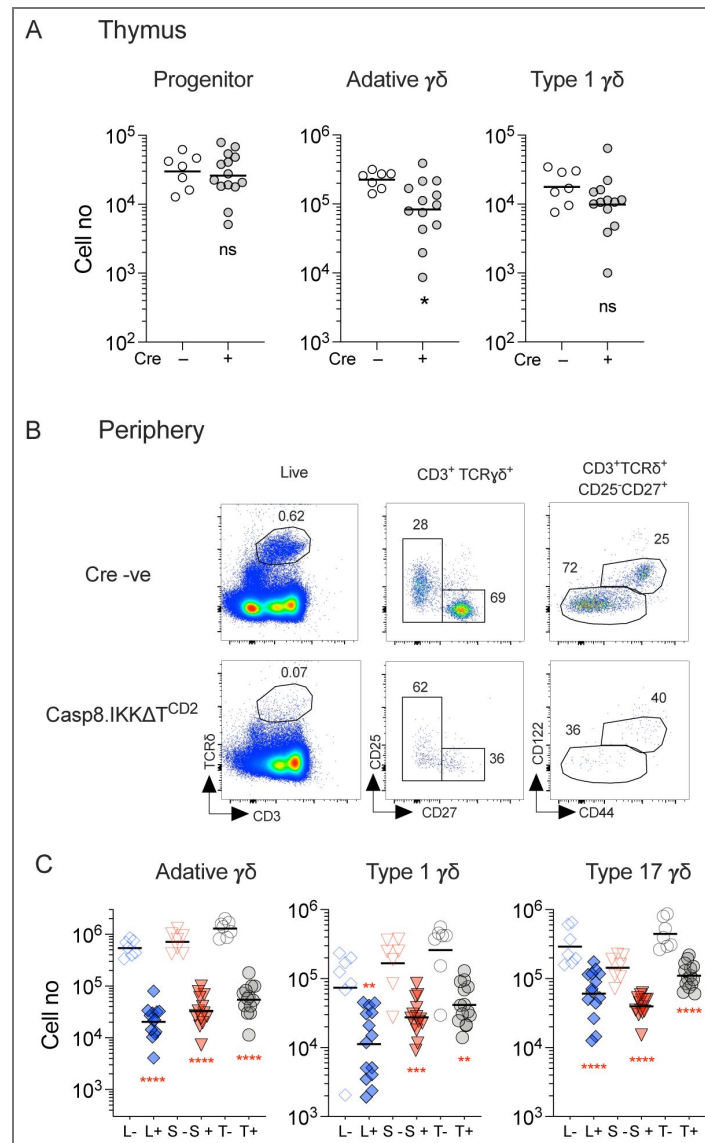
## Partial rescue of IKK-deficient peripheral $\gamma\delta$ T cells by kinase dead RIPK1

In  $\alpha\beta$  T cells, IKK kinase activity blocks CASPASE8-dependent apoptosis by directly phosphorylating RIPK1, which acts to repress its kinase activity that is required to induce CASPASE8-dependent cell death (Blanchett et al., 2022; Webb et al., 2019). The failure of CASPASE8 ablation to rescue peripheral  $\gamma\delta$  T cells potentially implied that IKK was not required to repress extrinsic cell death pathways in  $\gamma\delta$  T cells. However, while *Casp8* deletion would serve to protect cells from CASPASE8-dependent apoptosis, it could also result in triggering death instead by necroptosis. In T cells, extrinsic death pathways triggering CASPASE8-dependent apoptosis or necroptosis both depend on the kinase activity of RIPK1. Therefore, to test whether *Casp8* deletion might instead be triggering necroptosis in Casp8.IKK $\Delta$ T<sup>CD2</sup> mice, we analysed peripheral  $\gamma\delta$  T cell numbers in IKK $\Delta$ T<sup>CD2</sup> mice expressing kinase dead RIPK1<sup>D138N</sup>. As described earlier, IKK $\Delta$ T<sup>CD2</sup> mice are almost completely devoid of  $\gamma\delta$  T cells, except for type 17 cells that were present, albeit in reduced numbers (Fig. 3A-B). Analysing IKK $\Delta$ T<sup>CD2</sup> RIPK1<sup>D138N</sup> mice revealed a small but significant population of CD27<sup>+</sup>  $\gamma\delta$  T cells in the periphery of mice, that included both adaptive and type 1 subsets (Fig. 3A). In contrast, the reduction in type 17  $\gamma\delta$  T cells observed in IKK $\Delta$ T<sup>CD2</sup> mice was not restored in IKK $\Delta$ T<sup>CD2</sup> RIPK1<sup>D138N</sup> mice. These results suggest that  $\gamma\delta$  T cells do require IKK activity to repress RIPK1 dependent cell death pathways but that blocking this pathway alone is insufficient to restore peripheral  $\gamma\delta$  T cell compartment, and IKK is required to mediate additional functions to support peripheral  $\gamma\delta$  T cells. They also provided evidence that  $\gamma\delta$  T cells are susceptible to necroptosis, and that the failure of CASPASE8 ablation to rescue IKK deficient cells from cell death was due to a switch from apoptotic to necroptotic death in the absence of IKK expression.

## Adaptive and type 1 $\gamma\delta$ T cells are highly susceptible to necroptosis

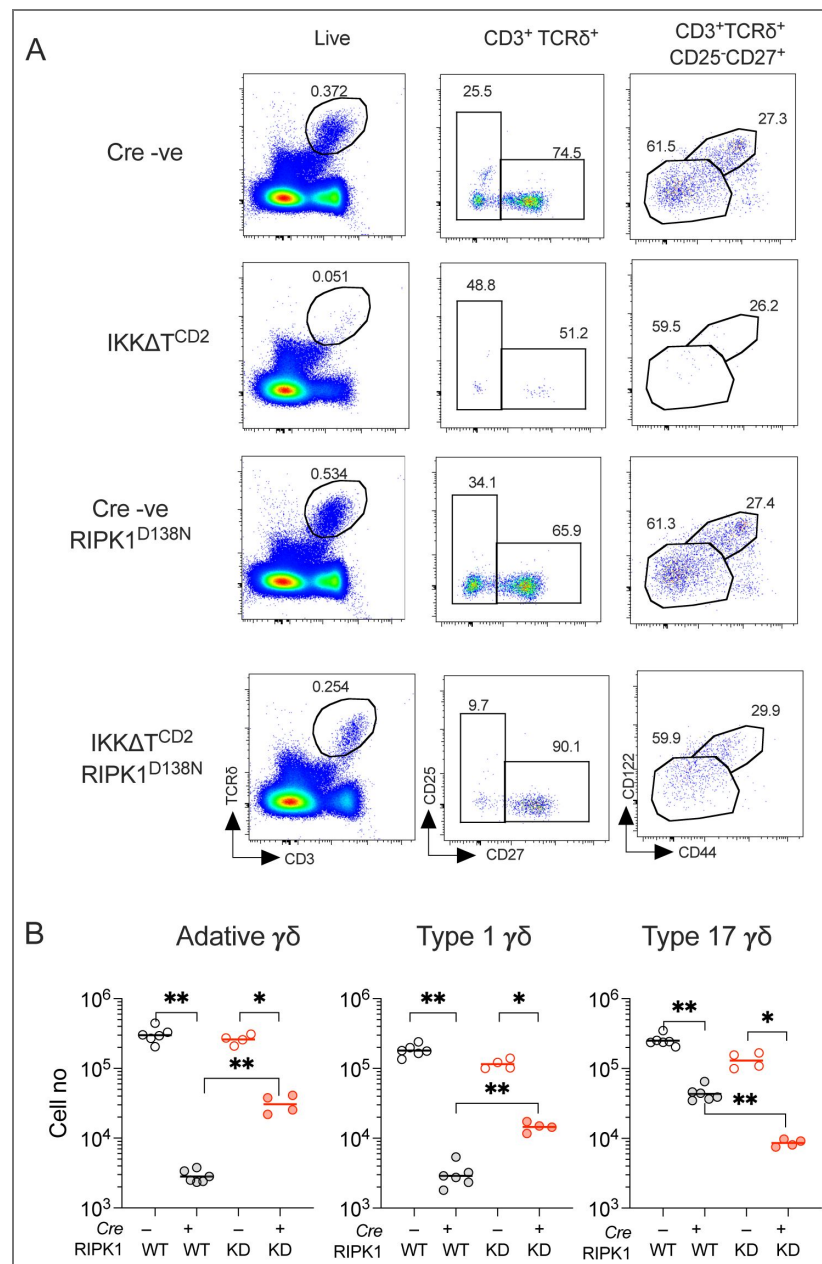
In order to directly test whether  $\gamma\delta$  T cells are susceptible to necroptosis, we next analysed mice in which *Casp8* alone was deleted in T cells, in Casp8 $\Delta$ T<sup>CD2</sup> mice. Analysing  $\gamma\delta$  T cell development in the thymus revealed normal representation (Fig. 4A) and numbers (Fig. 4B) of progenitor and newly generated adaptive and type 1  $\gamma\delta$  T cells. In the periphery, however, numbers of both





**Figure 2. CASPASE8 ablation does not rescue peripheral  $\gamma\delta$  T cell compartment of IKK $\Delta$ T<sup>CD2</sup> mice.**

Thymi, lymph nodes and spleen from Casp8.IKK $\Delta$ T<sup>CD2</sup> mice (n=13) and Cre -ve littermates (n=7) were enumerated and analysed by flow. (A) Scatter plots are of total cell numbers of the indicated subset recovered from thymi of Casp8.IKK $\Delta$ T<sup>CD2</sup> mice or Cre-ve littermates. (B) Representative flow plots from lymph nodes of the indicated mouse strains (rows), with the indicated electronic gates (columns). (C) Scatter plots are of total cell numbers recovered from lymph nodes (L), spleen (S) and total combined spleen and lymph nodes (T) of the indicated subset from Cre+ve (+) and Cre-ve (-) littermates. Horizontal bars are mean. Fractional representation of different  $\gamma\delta$  subsets in different organs is shown in Figure S2B. Data are pooled from multiple batches of mice analysed. Horizontal lines indicate mean. \*\* p<0.01, \*\*\*\* p<0.0001, Mann Whitney test.



**Figure 3.** Kinase dead RIPK1 mediates partial rescue of peripheral  $\gamma\delta$  T cell compartments of IKK $\Delta$ T<sup>CD2</sup> mice.

Lymph nodes and spleen from IKK $\Delta$ T<sup>CD2</sup> (n=6) IKK $\Delta$ T<sup>CD2</sup>RIPK1<sup>D138N</sup> mice (n=4) and Cre -ve littermates (n=6 and 4 respectively) were enumerated and analysed by flow. (A) Representative flow plots are of splenic cells from different strains (rows) with the indicated gates (columns). (B) Scatter plots are of total cell numbers of the indicated subset recovered from lymph nodes and spleen combined of the indicated IKK $\Delta$ T<sup>CD2</sup> strain expressing either WT or kinase dead (KD) RIPK1<sup>D138N</sup>. Data are pooled from two independent experiments. Horizontal lines indicate mean. \* p < 0.05, \*\* p < 0.01, Mann Whitney test.

adaptive and type 1  $\gamma\delta$  T cells were profoundly reduced with only a small population of  $CD27^+$   $\gamma\delta$  T cells remaining in lymph nodes and spleen (Fig. 4C-D). In contrast, no significant difference in the total number of type 17  $\gamma\delta$  T cells was apparent in the periphery, suggesting that CASPASE8 is not required for either development or maintenance of this subset (Fig. 4D).

To confirm that the loss of peripheral  $\gamma\delta$  T cells was due to necroptosis in the absence of *Casp8* expression, we generated *Casp8 $\Delta$ T<sup>CD2</sup> RIPK1<sup>D138N</sup>* mice expressing kinase dead RIPK1, to see if this would rescue cells from death. Analysing the periphery of this strain revealed normal representation (Fig. 5A) and numbers (Fig. 5B) of adaptive, type 1 and type 17  $\gamma\delta$  T cells, confirming that loss of adaptive and type 1 subsets in *Casp8 $\Delta$ T<sup>CD2</sup>* mice was the result of necroptosis in the absence of *Casp8* expression. We further confirmed that rescued cells in *Casp8 $\Delta$ T<sup>CD2</sup> RIPK1<sup>D138N</sup>* mice were the functional counterparts of the same subsets in Cre –ve controls by assessing cytokine production in vitro. IL-17A and IFN-gamma production was similar in extent and restricted to the corresponding type 17 and type 1 subsets (Fig. 5C).

## Alternative NF- $\kappa$ B activation is redundant for normal $\gamma\delta$ T cell homeostasis

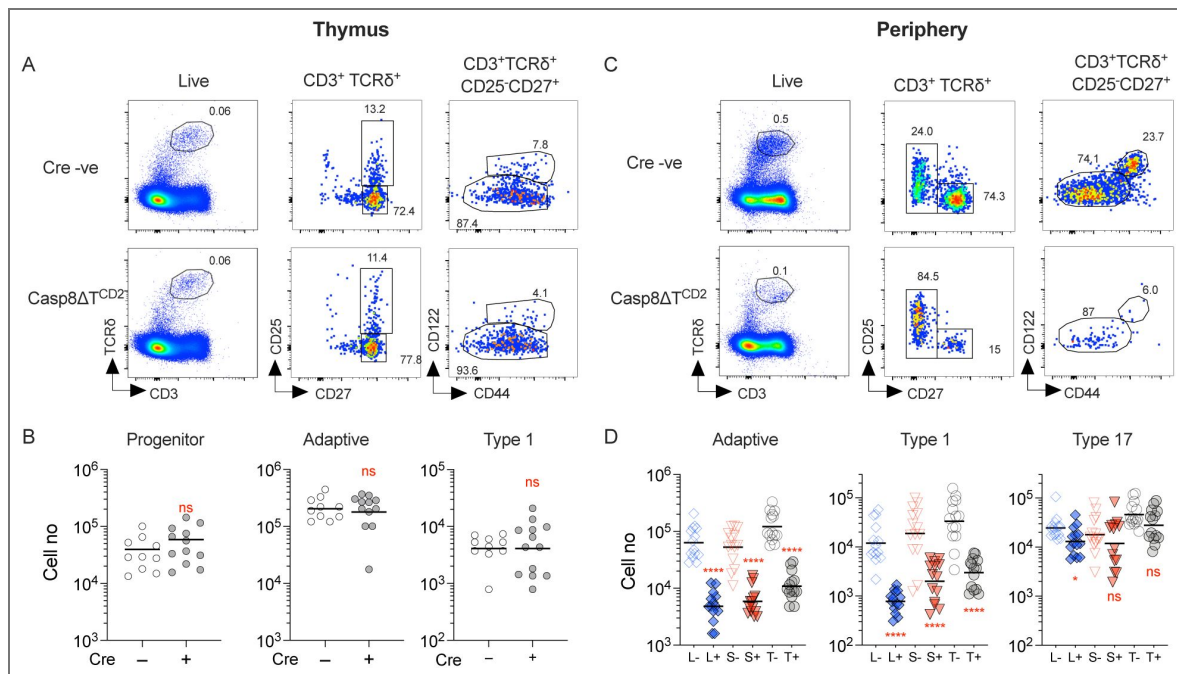
Our data demonstrated that generation of type 1 and long-term maintenance of adaptive  $\gamma\delta$  T cells was highly dependent upon IKK signalling. IKK kinase activity is required by T cells to both trigger NF- $\kappa$ B activation but also to directly block extrinsic cell death pathways by inhibiting RIPK1 activity (Blanchett et al., 2021). Kinase dead RIPK1 only achieved a modest rescue of peripheral  $\gamma\delta$  T cell numbers, suggesting that these cells also require the I $\kappa$ B kinase activity of IKK for their persistence, and that they also require NF- $\kappa$ B. We therefore dissected the specific role of NF- $\kappa$ B for  $\gamma\delta$  T cell homeostasis. IKK proteins regulate activation of both canonical NF- $\kappa$ B, mediated by RelA/p50 and cRel/p50 heterodimers, and non-canonical or alternative NF- $\kappa$ B, mediated by RelB/p52 dimers. A trimeric complex of IKK1, IKK2 and regulatory subunit NEMO is responsible for triggering canonical activation, while alternative NF- $\kappa$ B is triggered exclusively by homodimers of IKK1. In *IKK1 $\Delta$ T<sup>CD2</sup>* mice,  $\gamma\delta$  T cells are unable to activate either of these NF- $\kappa$ B pathways.

To determine if alternative NF- $\kappa$ B pathways contribute to the phenotype of *IKK1 $\Delta$ T<sup>CD2</sup>* mice, we analysed *IKK1 $\Delta$ T<sup>CD2</sup>* mice that lack IKK1 expression while retaining IKK2 expression. Canonical NF- $\kappa$ B activation in T cells is largely normal in the absence of IKK1, while alternative NF- $\kappa$ B is completely blocked (Lawrence, 2009). Analysing thymus from these mice showed that development of both adaptive and type 1  $\gamma\delta$  T cells was normal (Fig. 6A-B). Similarly, numbers and representation of adaptive, type 1 and type 17  $\gamma\delta$  T cells in the periphery were also normal in these mice as compared with Cre –ve littermates. Therefore, this suggests that alternative NF- $\kappa$ B pathways are redundant for both the development and persistence of  $\gamma\delta$  T cells.

## Canonical NF- $\kappa$ B is essential for maintenance of peripheral $\gamma\delta$ T cells

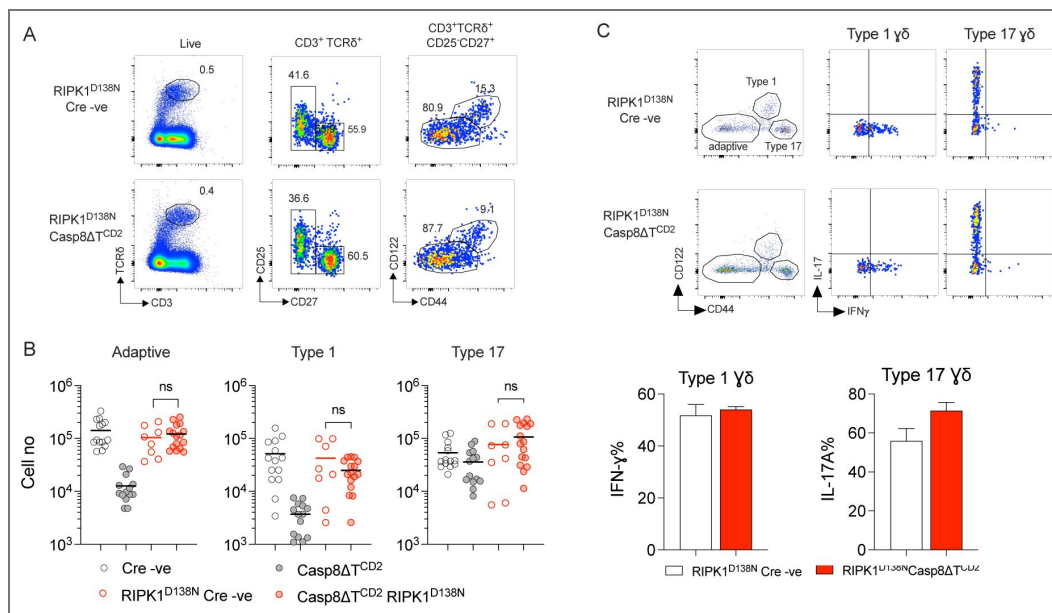
The phenotype of *IKK1 $\Delta$ T<sup>CD2</sup>* mice appeared to exclude a role for alternative NF- $\kappa$ B pathways in regulating  $\gamma\delta$  T cell homeostasis. Therefore, we next sought to dissect the role of canonical NF- $\kappa$ B pathways. The p50 subunit that derives from processing of NFKB1, lacks a transactivation domain, so while it can bind DNA, it cannot activate transcription alone. Therefore, combined ablation of cREL and RELA is sufficient to completely block canonical NF- $\kappa$ B pathways. Amongst conventional  $\alpha\beta$  T cells, cREL expression is redundant for naive T cells that can be supported by RELA alone, while memory phenotype T cells depend upon both cREL and RELA expression for their generation and/or persistence (Webb et al., 2019; Zheng et al., 2003). Therefore, we analysed mice whose T cells lacked expression of only RELA (*Rela $\Delta$ T<sup>CD2</sup>*), mice with germline *Nfkb1* deficiency (*Nfkb1<sup>-/-</sup>*), combined T cell deficiency of RELA and germline *Nfkb1* deficiency (*Rela $\Delta$ T<sup>CD2</sup> Nfkb1<sup>-/-</sup>*) and combined T cell specific ablation of both RELA and cREL (*Rela.Rel $\Delta$ T<sup>CD2</sup>*). We first analysed the thymus of these strains to assess the development of adaptive and type 1  $\gamma\delta$  T cells. Numbers of progenitor and adaptive  $\gamma\delta$  T cells appeared largely normal (Fig. 7A), while there was some statistical evidence of a modest (~2 fold) reduction in absolute numbers of both





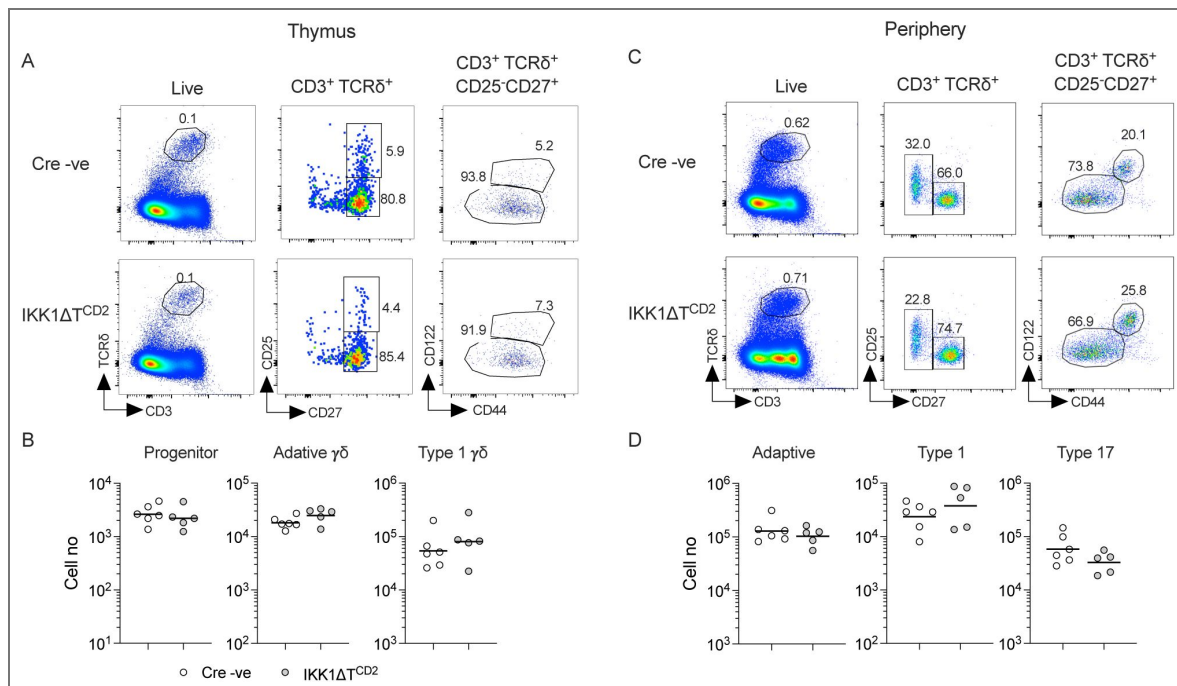
**Figure 4. CASPASE8 expression is critical for long-term survival of peripheral  $\gamma\delta$  T cells.**

Thymi, lymph nodes and spleen from Casp8ΔT<sup>CD2</sup> mice (n=13) and Cre -ve littermates (n=10) were enumerated and analysed by flow. (A) Representative flow plots are of thymocytes with the indicated gates, showing gates used to identify progenitor, adaptive and type 1 subsets. (B) Scatter plots are of total cell numbers of the indicated subset from thymus of Casp8ΔT<sup>CD2</sup> mice (+) and Cre -ve littermates (-). (C) Representative flow plots illustrate gating strategy to identify adaptive, type 1 and type 17 subsets in lymph nodes (shown) and spleen from the indicated strains. (D) Scatter plots are of total cell numbers recovered from lymph nodes (L), spleen (S) and total combined spleen and lymph nodes (T), of the indicated subsets from Cre +ve (+) and Cre -ve (-) littermates. Horizontal bars are means. Fractional representation of different  $\gamma\delta$  subsets in different organs is shown in Figure S2C. Data are pooled from five independent experiments. Horizontal lines indicate mean. n.s - not significant, \*\*\* p<0.0001, Mann Whitney test.



**Figure 5. Kinase dead RIPK1 fully restores the peripheral  $\gamma\delta$  T cell compartments of Casp8 $\Delta$ T<sup>CD2</sup> mice.**

Lymph nodes and spleen from Casp8 $\Delta$ T<sup>CD2</sup> RIPK1<sup>D138N</sup> mice (n=16) and Cre -ve RIPK1<sup>D138N</sup> littermates (n=8) were enumerated and analysed by flow. (A) Representative flow plots are of lymph nodes from Casp8 $\Delta$ T<sup>CD2</sup> RIPK1<sup>D138N</sup> mice and Cre -ve RIPK1<sup>D138N</sup> littermates with the indicated gates applied (columns). (B) Scatter plots are of total cell numbers of the indicated subsets from both lymph nodes and spleen combined from Casp8 $\Delta$ T<sup>CD2</sup> RIPK1<sup>D138N</sup> mice and Cre -ve RIPK1<sup>D138N</sup> littermates (red symbols) and cell numbers from Casp8 $\Delta$ T<sup>CD2</sup> RIPK1<sup>WT</sup> strains described in figure 4, for direct comparison. (C) Lymph node cells were stimulated in vitro with calcium ionophore and phorbol esters for 4 hours with brefeldin A, and then analysed for the indicated intracellular cytokines. Representative flow plots illustrate gating strategy to identify adaptive, type 1 and type 17 subsets on the basis of CD44 and CD122 expression. Quad gates for cytokine detection were set against negative controls of matched unstimulated cells. Bar charts are of total % of cells stained for IFN-gamma or IL-17A. Data are pooled from multiple batches of mice analysed (A-B) or are the pooled from 4 independent experiments (C). n.s - not significant, Mann Whitney test.



**Figure 6. Alternative NF-κB signalling is redundant for generation and maintenance of  $\gamma\delta$  T cell compartments.**

Thymi, lymph nodes and spleen from  $IKK1\Delta T^{CD2}$  mice (n=5) and Cre -ve littermates (n=6) were enumerated and analysed by flow. (A) Representative flow plots are of thymocytes from  $IKK1\Delta T^{CD2}$  mice and Cre -ve littermate, showing gates used to identify progenitor, adaptive and type 1 subsets intrathymically. (B) Scatter plots are of total cell numbers of the indicated subset from thymus. (C) Representative flow plots illustrate gating strategy to identify adaptive, type 1 and type 17 subsets in lymph nodes (shown) and spleen. (D) Scatter plots are of total cell numbers recovered from both spleen and lymph nodes of the indicated subset. Data are pooled from two independent experiments.

these subsets in some of the REL deficient strains. This was in contrast to the absence of a similar phenotype in  $IKK\Delta T^{CD2}$  mice (Fig. 1 [↗](#)). We therefore also compared representation of progenitors  $\gamma\delta$  T cells with DN3 cell numbers, that reflect upstream progenitor pool. This ratio was similar across strains (Fig. 7A [↗](#)), suggesting that the observed modest reductions were not cell intrinsic but may rather reflect differences in overall thymic cellularity from these strains. In contrast, analysing representation and numbers of type 1  $\gamma\delta$  T cells revealed similar reductions as observed in  $IKK1\Delta T^{CD2}$  mice, suggesting that development of this subset is indeed dependent upon canonical NF- $\kappa$ B. Furthermore, analysing the impact of different REL subunit ablations suggested that development of this subset was highly dependent upon NF- $\kappa$ B, since even ablation of RELA alone was sufficient to reduce the numbers of newly generated cells in the thymus, while combined ablation of both RELA and cREL resulted in the greatest reduction in numbers of type 1  $\gamma\delta$  T cells (Fig. 7A [↗](#)).

Finally, we analysed the peripheral compartments of different REL deficient strains to determine the specific requirements for NF- $\kappa$ B by different subsets. In mice lacking RELA alone, RELA/NF $\kappa$ B1 or RELA/cREL, both type 1 and type 17  $\gamma\delta$  T cells were substantially reduced (Fig. 7B [↗](#)). The reduction in numbers of type 1  $\gamma\delta$  T cells in the periphery of these mice mirrored reductions observed in the thymus, further reinforcing the view that NF- $\kappa$ B signalling was essential for their normal thymic generation. The reduction of type 17  $\gamma\delta$  T cells suggests that NF- $\kappa$ B is also necessary for the development and/or maintenance of this population. Type 1 and type 17  $\gamma\delta$  T cells were similarly reduced in all three REL deficient strains, suggesting a strong dependence upon NF- $\kappa$ B. Numbers of peripheral adaptive  $\gamma\delta$  T cells were also profoundly reduced in RELA/ cREL deficient mice, but there was evidence for redundancy between subunits, since specific ablation of RELA alone was well tolerated (Fig. 7B [↗](#)). Overall, these data show that maintenance of replete peripheral  $\gamma\delta$  T cell compartments is highly dependent on tonic NF- $\kappa$ B signals either for development or maintenance of different subsets.

## Discussion

The IKK complex is a critical regulator of cell death, differentiation and inflammation in multiple cell types, including  $\alpha\beta$  T cells. In the present study, we sought to understand whether IKK signalling was also important for the development and maintenance of  $\gamma\delta$  T cells, which are implicated as early responders in a host of different inflammatory settings. We found that IKK was critical for establishing a mature  $\gamma\delta$  T cell compartment, but found evidence for contrasting roles and mechanisms of downstream pathways for the development and maintenance of different subsets.

The inhibitor of  $\kappa$ B kinase activity of the IKK complex is critical for activating NF- $\kappa$ B signalling, so we were able to demonstrate the essential role of this transcription factor by two independent means - either ablating the upstream IKK complex or direct ablation of canonical REL subunits. Numbers of type 1  $\gamma\delta$  T cells in the thymus were substantially reduced in both settings, suggesting that generation of this pre-differentiated subset requires NF- $\kappa$ B signalling to support their ontogeny. In contrast, NF- $\kappa$ B signalling was redundant for both specification of progenitors to a  $\gamma\delta$  T cell lineage and generation of adaptive  $\gamma\delta$  T cells, as numbers of progenitors and adaptive  $\gamma\delta$  T cells were largely normal in both REL and IKK deficient strains. Strikingly, however, mature adaptive  $\gamma\delta$  T cells were largely absent in both REL and IKK deficient mice, revealing that NF- $\kappa$ B is critical for maintenance of mature  $\gamma\delta$  T cells. Assessing the role of different REL subunits also revealed distinct requirements for development vs maintenance that mirrored preferences observed in  $\alpha\beta$  T cells. Generation of effector  $\alpha\beta$  T cells is highly dependent upon NF- $\kappa$ B, since loss of cREL alone is sufficient to block development of  $\alpha\beta$  T cell memory (Webb et al., 2019 [↗](#); Zheng et al., 2003 [↗](#)).

Similarly, we found that loss of RELA alone was sufficient to impair development of type 1  $\gamma\delta$  T cells, suggesting the cREL and p50 alone are insufficient for their development. Furthermore, in naive  $\alpha\beta$  T cell, there is evidence of redundancy between REL subunits. Expression of cREL and p50 in the absence of RELA is sufficient to maintain naive  $\alpha\beta$  T cells, while expression of cREL alone, in RELA/p50 double knockouts is not (Webb et al., 2019 [↗](#)). We also found this to be the



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exact case for adaptive  $\gamma\delta$  T cells. These subtly distinct requirements for NF- $\kappa$ B probably reflect both distinct biological processes involved, ontogeny vs maintenance, but also potentially the receptors involved in triggering signalling. TNFRSF members appear to be implicated in triggering survival signalling in naive  $\alpha\beta$  T cells (Silva et al., 2014 [\[14\]](#)) while generation of differentiated  $\alpha\beta$  T effectors is dependent on strong agonist triggers from TCR and also TNFRSF members (Layzell et al., 2024 [\[15\]](#)). This is also true of Foxp3<sup>+</sup> regulatory T cells, that require both RELA and cREL for their development (Isomura et al., 2009 [\[16\]](#); Messina et al., 2016 [\[17\]](#)), and are induced by a combination of both agonist TCR and TNFRSF signalling, amongst other receptors.

In  $\alpha\beta$  T cells, IKK also plays a crucial role in controlling cell survival directly by its repressive activity upon the cell death regulator, RIPK1. Developing thymocytes do not require NF- $\kappa$ B signalling for development but do become exquisitely sensitive to TNF-induced cell death in the absence of IKK (Webb et al., 2019 [\[18\]](#)), while mature naive T cells require IKK expression to both repress RIPK1 and activate an NF- $\kappa$ B dependent survival signal. The balance between control of NF- $\kappa$ B and blocking cell death by IKK appears different in  $\gamma\delta$  T cells, as does the control of cell death processes themselves. Neither  $\alpha\beta$  nor  $\gamma\delta$  thymocytes undergo necroptosis in the absence of CASPASE8, while, in stark contrast to resting  $\alpha\beta$  T cells, adaptive and type 1  $\gamma\delta$  T cells in the periphery were exquisitely sensitive to the induction of necroptosis in the absence of CASPASE8. This appears to be determined in part by expression of MLKL that is present in  $\gamma\delta$  T cells but absent from resting  $\alpha\beta$  T cells (ImmGen browser enquiry (Heng and Painter, 2008 [\[19\]](#))). Using kinase dead RIPK1, it was possible to test the importance of regulation of RIPK1-dependent cell death pathways by IKK, since this RIPK1 mutant blocks both necroptosis and apoptosis, while CASPASE8 ablation served to block apoptosis but also redirected cell death processes to a necroptotic modality. IKK $\Delta$ T<sup>CD2</sup> mice expressing kinase dead RIPK1 did exhibit some significant, albeit incomplete, rescue of the peripheral T cell compartment. The phenotype of REL-ablated mice confirmed that IKK is required to trigger an NF- $\kappa$ B dependent survival programme. The relative importance of IKK induced NF- $\kappa$ B vs IKK mediated repression of RIPK1 for the maintenance of peripheral  $\gamma\delta$  T cells is not easy to disentangle. The extent of peripheral  $\gamma\delta$  T loss in the absence of IKK was substantially greater than observed in REL deficient strains. Kinase dead RIPK1 restored peripheral  $\gamma\delta$  T cell levels to similar to those observed in REL deficient mice, suggesting that the impact of IKK deficiency is indeed a compound effect of losing both IKK induced NF- $\kappa$ B activity and loss of IKK dependent repression of RIPK1. It would be interesting to test the impact of losing IKK dependent control of RIPK1 in the context of normal NF- $\kappa$ B activation upon peripheral  $\gamma\delta$  T cell compartment.

In order to target gene deletion prior to  $\gamma\delta$  T cell commitment, we employed the use of huCD2<sup>iCre</sup> that is expressed from common lymphoid progenitors. This will inevitably also target deletion in  $\alpha\beta$  T cells, the consequences of which have already been extensively characterised for IKK, REL and CASPASE8 genes (Islam et al., 2025 [\[20\]](#); Webb et al., 2019 [\[18\]](#); Webb et al., 2016 [\[17\]](#)). This does raise the question of whether alterations to the  $\alpha\beta$  T cell compartment effect the phenotypes we describe amongst  $\gamma\delta$  T cells. Studies of TCR $\alpha$  knockout mice confirm that development and maintenance of  $\gamma\delta$  T cells does not depend upon the presence of  $\alpha\beta$  T cells (Hayday and Tigelaar, 2003 [\[21\]](#); Philpott et al., 1992 [\[22\]](#)). However, in an absence of peripheral  $\alpha\beta$  T cells, as is the case in IKK $\Delta$ T<sup>CD2</sup> mice, there will be reduced competition for cytokines that both lineages rely upon, such as IL-7 and IL-15 (French et al., 2005 [\[23\]](#)). Potential alterations to homeostatic niches did not appear to impact our results however. The loss of  $\gamma\delta$  T cells in the strains we investigated here occurred both in the presence and absence of  $\alpha\beta$  T cells. We observed normal thymic development of  $\gamma\delta$  T cells but their complete absence from peripheral compartments in both IKK $\Delta$ T<sup>CD2</sup> and Casp8.IKK $\Delta$ T<sup>CD2</sup> strains, in spite of the fact that development and peripheral population of  $\alpha\beta$  T cell compartment is completely blocked in IKK $\Delta$ T<sup>CD2</sup> mice but restored in Casp8.IKK $\Delta$ T<sup>CD2</sup> mice. Therefore, an absence of competition for IL-7 and IL-15 in IKK $\Delta$ T<sup>CD2</sup> mice is not sufficient to overcome survival defects of  $\gamma\delta$  T cells we observed in these strains and the defects in the  $\gamma\delta$  T cell compartment do appear cell intrinsic.

Type 17  $\gamma\delta$  T cells are primarily generated during embryonic stages of development (Munoz-Ruiz et al., 2017 [DOI](#)), and so we could not directly observe their intrathymic generation in adult mice. As such, the presence or absence of cells in adult mice could reflect either developmental processes and/or their defective survival. Nevertheless, there were still some useful conclusions we could draw. The maintenance of the mature type 17  $\gamma\delta$  T cells does depend on NF- $\kappa$ B signalling. Peripheral numbers of type 17  $\gamma\delta$  T cells were reduced by around a two-thirds in both IKK and REL deficient strains. However, in contrast to other subsets, there was no evidence that kinase dead RIPK1 mediated any rescue of type 17 subset, and they did not appear to be sensitive to necroptosis in the absence of CASPASE8. Their numbers were also not restored in Casp8.IKK $\Delta$ T<sup>CD2</sup> mice, strongly suggesting that the sole defect in IKK deficient strains was the absence of NF- $\kappa$ B activation. An important caveat here is that REL and IKK deficient strains also lack effector  $\alpha\beta$  T cell populations in the absence of NF- $\kappa$ B activation (Webb et al., 2019 [DOI](#)). There are reports of “trans-conditioning”, whereby development of type 17  $\gamma\delta$  T cells is promoted by type 17  $\alpha\beta$  T cells (Do et al., 2012 [DOI](#)). Therefore, it is possible that a failure of trans-conditioning by  $\alpha\beta$  type 17 T cells is a contributing factor in REL and IKK deficient strains. However, our findings are also consistent with other studies that show that RelA is involved in development of type 17  $\gamma\delta$  T cells (Powolny-Budnicka et al., 2011 [DOI](#)). The same study also implicated RelB transcription factor in the development of type 17  $\gamma\delta$  T cells. This appears at odds with our findings of normal development of these cells in IKK1 deficient mice, since IKK1 homodimers mediate activation of the alternative NF- $\kappa$ B pathway. It is possible that type 17  $\gamma\delta$  T cells are still RelB dependent in IKK1 $\Delta$ T<sup>CD2</sup> mice but that there is redundancy between IKK subunits to activate RelB containing NF- $\kappa$ B dimers, permitting development of these cells. The stark contrast in sensitivity of type 17 and other subsets to necroptosis in the absence of CASPASE8 was surprising, as they all appear to express MLKL (ImmGen browser enquiry (Heng and Painter, 2008 [DOI](#))). It is possible this may reflect altered cell death control by the distinct fetal liver derived progenitors from which type 17 cells are derived. Whichever the case, if confirmed, it would be of interest to identify the mechanism by which these cells are resistant to necroptosis, since they appear to express the basic elements required to form the necrosome.

In conclusion, our study reveals the complexity of regulation of inflammatory and cell death pathways in T cells of the adaptive immune system. In other cell types and tissues, IKK mediates well defined pro-inflammatory and pro-survival functions. In contrast, in T cells, the functions of IKK and how cell death pathways are regulated varies with differentiation state and lineage. IKK dependent repression of RIPK1 is critical for thymic development of  $\alpha\beta$  T cells, but is redundant for thymic development in  $\gamma\delta$  T cells. In common with  $\alpha\beta$  T cells, mature  $\gamma\delta$  T cells require IKK to both repress RIPK1 dependent cell death and activate NF- $\kappa$ B survival, with perhaps a more dominant requirement for the latter function. We found evidence that intrathymic development of type 17  $\gamma\delta$  T cells did depend upon NF- $\kappa$ B. Their development is thought to be driven by strong “agonistic” TCR signalling, and it is perhaps significant that other agonist selected populations, such as NK T cells and Foxp3<sup>+</sup> regulatory T cells are also dependent upon NF- $\kappa$ B signalling (Isomura et al., 2009 [DOI](#); Sivakumar et al., 2003 [DOI](#)), implying a conserved developmental mechanisms govern generation of such agonist selected populations in the thymus. It is also interesting to note that like  $\gamma\delta$  T cells, NK T cells and Treg are also susceptible to necroptosis in the absence of CASPASE8 expression (Islam et al., 2025 [DOI](#); Teh et al., 2022 [DOI](#)), in contrast to resting  $\alpha\beta$  T cells are largely resistant. This does contrast with adaptive  $\gamma\delta$  T cells that do become exquisitely sensitive to necroptosis induction as soon as they enter the periphery. This difference may reflect the role of  $\gamma\delta$  T cells as first line early responders in immune responses, that may expose them to microbial interference of cell death pathways, that necroptosis has evolved to resist. Whichever is the case, a detailed understanding of these pathways will be critical to understand the full impact of therapies that target these potent inflammatory pathways, and that will have diverse and complex impact on adaptive immunity.

## Materials and methods

### Mice

Mice with the following mutations were used in this study; B6.129-Casp8<sup>tm1Hed/J</sup> (*Casp8*<sup>fx</sup>), *Ikbkbtm2Mka* (Li et al., 2003) (*Ikbkb*<sup>fx</sup>), *Chukrm1Mpa* (Gareus et al., 2007) (*Chuk*<sup>fx</sup>), *Relatm1Asba* (Steinbrecher et al., 2008) (*Rela*<sup>fx</sup>), B6.129S1-Reltm1Ukl/J (Heise et al., 2014) (*Rel*<sup>fx</sup>), *Nfkb1tm1Bal* (*Nfkb1*<sup>null</sup>), *Cre* transgenes expressed under the control of the human CD2, B6.Cg-Tg(CD2-icre)4Kio/J (*huCD2*<sup>iCre</sup>) (de Boer et al., 2003) mice with a D138N mutation in *Ripk1*, B6.129-Ripk1<sup>tm1Geno/J</sup> (*RIPK1*<sup>D138N</sup>) (Newton et al., 2014). The following strains were bred using these alleles for this study; *Chuk*<sup>fx/fx</sup> *Ikbkb*<sup>fx/fx</sup> *huCD2*<sup>iCre</sup> (*IKKΔT*<sup>CD2</sup>), *Chuk*<sup>fx/fx</sup> *huCD2*<sup>iCre</sup> (*IKK1ΔT*<sup>CD2</sup>), *Chuk*<sup>fx/fx</sup> *Ikbkb*<sup>fx/fx</sup> *Casp8*<sup>fx/fx</sup> *huCD2*<sup>iCre</sup> (*Casp8.IKKΔT*<sup>CD2</sup>), *Chuk*<sup>fx/fx</sup> *Ikbkb*<sup>fx/fx</sup> *huCD2*<sup>iCre</sup> *Ripk1*<sup>D138N</sup> (*IKKΔT*<sup>CD2</sup> *RIPK1*<sup>D138N</sup>), *Casp8*<sup>fx/fx</sup> *huCD2*<sup>iCre</sup> (*Casp8ΔT*<sup>CD2</sup>), *Casp8*<sup>fx/fx</sup> *huCD2*<sup>iCre</sup> *Ripk1*<sup>D138N</sup> (*Casp8ΔT*<sup>CD2</sup> *RIPK1*<sup>D138N</sup>), *Rela*<sup>fx</sup> *huCD2*<sup>iCre</sup> (*RelaΔT*<sup>CD2</sup>), *Rela*<sup>fx/fx</sup> *huCD2*<sup>iCre</sup> (*Rela.RelΔT*<sup>CD2</sup>), *Rela*<sup>fx/fx</sup> *huCD2*<sup>iCre</sup> *Nfkb1*<sup>-/-</sup> (*RelaΔT*<sup>CD2</sup> *Nfkb1*<sup>-/-</sup>). All mice were bred in the Comparative Biology Unit of the Royal Free UCL campus and at Charles River laboratories, Manston, UK. Animal experiments were performed according to institutional guidelines and Home Office regulations under project licence PP2330953.

### Flow cytometry and electronic gating strategies

Flow cytometric analysis was performed with 10 × 10<sup>6</sup> thymocytes, 5 × 10<sup>6</sup> lymph node (LN) or spleen cells. Cell concentrations of thymocytes, lymph node and spleen cells were determined with a Scharf Instruments Casy Counter. Cells were incubated with saturating concentrations of antibodies in 100 μl of Dulbecco's phosphate-buffered saline (PBS) containing bovine serum albumin (BSA, 0.1%) for 1 hour at 4°C followed by two washes in PBS-BSA. Panels used the following mAb: BV421-conjugated antibody against CD27 (Biolegend), PE-conjugated antibody against CD127 (ThermoFisher Scientific), BV785-conjugated CD44 antibody (Biolegend), BUV395-conjugated antibody against CD25 (BD horizon), PE-Cy7-conjugated antibody against CD122 (Biolegend), PerCP-cy5.5-conjugated antibody against CD3 (eBioscience), APC-conjugated antibody against TCRδ (eBioscience), FITC-conjugated antibody against IFN-γ (BD Biosciences), PE-Cy7-conjugated antibody against IL-17A (eBioscience). Cell viability was determined using LIVE/DEAD cell stain kit (Invitrogen Molecular Probes), following the manufacturer's protocol. multi-color flow cytometric staining was analyzed on a LSRFortessa (Becton Dickinson) instrument, and data analysis and color compensations were performed with FlowJo V10 software (TreeStar).

The following gating strategies were used : γδ progenitor T cells - CD3<sup>+</sup> TCRδ<sup>hi</sup> CD25<sup>+</sup> CD27<sup>+</sup>, adaptive γδ T cells - CD3<sup>+</sup> TCRδ<sup>+</sup> CD25<sup>-</sup> CD27<sup>+</sup> CD44<sup>lo</sup> CD122<sup>lo</sup>, type 1 γδ T cells - CD3<sup>+</sup> TCRδ<sup>+</sup> CD25<sup>-</sup> CD27<sup>+</sup> CD44<sup>+</sup> CD122<sup>+</sup>, type 17 γδ T cells - CD3<sup>+</sup> TCRγδ<sup>+</sup> CD25<sup>-</sup> CD27<sup>+</sup> respectively.

### Intracellular cytokine staining

LN T cells (cervical, auxiliary, brachial, inguinal and mesenteric) were cultured at 37°C with 5% CO<sub>2</sub> in RPMI-1640 (Gibco, Invitrogen Corporation, CA) supplemented with 10% (v/v) fetal bovine serum (FBS) (Gibco Invitrogen), 0.1% (v/v) 2-mercaptoethanol βME (Sigma Aldrich) and 1% (v/v) penicillin-streptomycin (Gibco Invitrogen) (RPMI-10). Approximately 5 × 10<sup>6</sup> LN cells were stimulated with a protein transport inhibitor cocktail (Brefeldin and Monensin) at 12.6 μM concentration and cell stimulation cocktail (PMA and Ionomycin) at 1.42 μM concentration resuspended in RPMI. The stimulated cells were incubated for 4 hours, followed by the cell surface and intracellular cytokine staining.

### Statistics

Statistical analysis and bar charts were performed using Graphpad Prism 9.2. Column data compared by two-tailed T-test (non-parametric) Mann-Witney t-test. Results were denoted as ns = nonsignificant, \* p<0.05, \*\* p<0.01, \*\*\* p<0.001 and \*\*\*\* p<0.0001.

## Data availability

Data available upon request.

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## Additional files

[Supplemental figures](#) 

## Additional information

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## Peer reviews

### Reviewer #1 (Public review):

Summary:



The NF- $\kappa$ B signaling pathway plays a critical role in the development and survival of conventional alpha beta T cells. Gamma delta T cells are evolutionarily conserved T cells that occupy a unique niche in the host immune system and that develop and function in a manner distinct from conventional alpha beta T cells. Specifically, unlike the case for conventional alpha beta T cells, a large portion of gamma delta T cells acquire functionality during thymic development, after which they emigrate from the thymus and populate a variety of mucosal tissues. Exactly how gamma delta T cells are functionally programmed remains unclear. In this manuscript, Islam et al., use a wide variety of mouse genetic models to examine the influence of the NF- $\kappa$ B signaling pathway on gamma delta T cell development and survival. They find that the inhibitor of kappa B kinase complex (IKK) is critical to the development of gamma delta T1 subsets, but not adaptive/naïve gamma delta T cells. In contrast, IKK-dependent NF- $\kappa$ B activation is required for their long-term survival. They find that caspase 8-deficiency renders gamma delta T cells sensitive to RIPK1-mediated necroptosis and they conclude that IKK repression of RIPK1 is required for the long-term survival of gamma delta T1 and adaptive/naïve gamma delta T cells subsets. These data will be invaluable in comparing and contrasting the signaling pathways critical for the development/survival of both alpha beta and gamma delta T cells.

Comments on revisions:

The word adaptive is misspelt throughout most figures.

<https://doi.org/10.7554/eLife.108940.2.sa2>

## Reviewer #2 (Public review):

This study presents a comprehensive genetic dissection of the role of IKK signaling in the development and maintenance of lymphoid gd T cells. By employing a variety of conditional and mutant mouse models, the authors demonstrate that IKK-dependent NF- $\kappa$ B activation is essential for the generation of type 1 gd T cells, while adaptive gd T cells require this pathway primarily for long-term survival. The use of multiple complementary genetic strategies, including IKK deletion and modulation of RIPK1 and CASPASE8 activity, provides robust mechanistic insight into subset-specific regulation of gd T cell homeostasis. Overall, the study provides mechanistic insight for IKK-dependent regulation of gd T cell development and peripheral maintenance.

Comments on revisions:

Thank you for your comments and clarifications.

<https://doi.org/10.7554/eLife.108940.2.sa1>

## Author response:

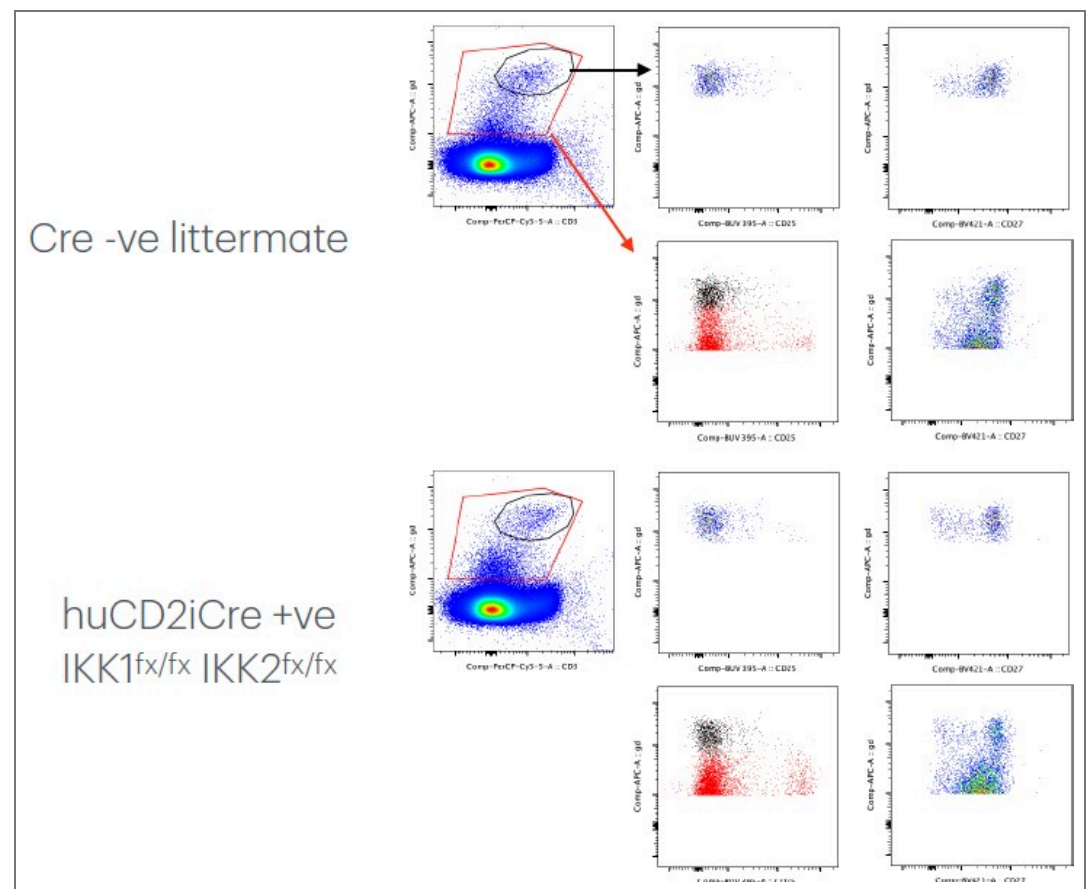
The following is the authors' response to the original reviews

### Public Reviews:

#### Reviewer #1 (Public review):

(1) The authors appear to be excluding a significant fraction of the TCR<sup>low</sup> gamma delta T cells from their analysis in Figure 1A. Since this population is generally enriched in CD25<sup>+</sup> gamma delta T cells, this gating strategy could significantly impact their analysis due to the exclusion of progenitor gamma delta T cell populations.

We were cautious in our gating strategy since the  $\text{TCR}\delta^+ \text{CD}3\epsilon^+$  subset is rather small and so low signal/background noise ratio can be an issue if the gates used are too broad/generous. There is some inevitable low level background staining with the  $\text{TCR}\delta$  that sits just above the bulk of the negative population and is  $\text{CD}3\epsilon^-$ . Although this background represents a tiny fraction of total cells, we were wary of gate contamination into our  $\text{TCR}\delta^+ \text{CD}3\epsilon^+$  subset and we wanted a gating strategy that could be applied across other organs too. We do not, however, believe this conservative strategy is impacting on measurements progenitor numbers across strains or our conclusions, since the size of this progenitor population in the various  $\text{IKK}\Delta\text{T}^{\text{CD}2}$  and  $\text{Casp}8\Delta\text{T}^{\text{CD}2}$  strains was never impacted by the mutations. But to reassure the reviewer, we show our conservative gate as compared with a very broad  $\text{TCR}\delta$  gate and see we are not missing a substantial population of  $\text{CD}25^+$  cells just below our gate. This also helps illustrate how close the background from the  $\text{CD}27^{\text{int}}$  expressing  $\alpha\beta$  thymocytes (right column) comes to the  $\text{TCR}\delta^+ \text{CD}3^+$  gate and the importance of tight lineage gating.



Author response image 1.

(2) The overall phenotype of the  $\text{IKK}\Delta\text{T}^{\text{CD}2}$  mice is not described in any great detail. For example, it is not clear if these mice possess altered thymocyte or peripheral T cell populations beyond that of gamma delta T cells.

Given that gamma delta T cell development has been demonstrated to be influenced by gamma delta T cells (i.e, trans-conditioning), this information could have aided in the interpretation of the data.

Apologies for not being clearer on this point. We have studied conventional  $\alpha\beta$  T cell development in these strains in considerable detail, and these studies are published and discussed in some detail in the introduction in paragraph 3 on page 3-4 and in cited references Schmidt-Suppran et al 2004, Silva et al 2014, Xing et al 2016, Webb et al 2019, Carty et al 2023. These detail how IKK expression is critical for thymic development of  $\alpha\beta$  T cells and their peripheral survival, and dissects the role of NF- $\kappa$ B activation and cell death regulation by IKK. However, we now add new discussion (page 11-12) that considers the potential impact of altered  $\alpha\beta$  T cell development in the strains used for this study.

We agree that trans-conditioning is also an important consideration, since CD4 TH17 T cells can enhance type 17  $\gamma\delta$  T cell development (10.1038/icb.2011.50). This is of relevance to the limited conclusions we draw concerning type 17  $\gamma\delta$  T cells. The REL and IKK deficient strains do lack effector populations, including type 17  $\alpha\beta$  T cells, so it is possible that the absence of type 17  $\alpha\beta$  T cells in these strains does contribute to the modest impact of IKK deletion in the type 17  $\gamma\delta$  subset. We now highlight this information and discuss in the manuscript (page 11-12).

*Related to this, it would have been helpful if the authors provided a comparison of the frequencies of each of the relevant subsets, in addition to the numbers.*

We now provide both the absolute frequencies of different  $\gamma\delta$  subsets and their relative frequencies to one another, as supplementary figure 2. We still believe assessing absolute numbers is the gold standard, since the differential impact of gene deletions on the  $\alpha\beta$  T cell compartments in different strains will effect whether or not  $\alpha\beta$  T cells are present, and therefore overall representation of  $\gamma\delta$  T cells can vary considerably between strains. Hence, absolute numbers are more reliable measure of cell abundance.

*(3) The manner in which the peripheral gamma delta T cell compartment was analyzed is somewhat unclear. The authors appear to have assessed both spleen and lymph node separately. The authors show representative data from only one of these organs (usually the lymph node) and show one analysis of peripheral gamma delta T cell numbers, where they appear to have summed up the individual spleen and lymph node gamma delta T cell counts. Since gamma deltaT17 and gamma deltaT1 are distributed somewhat differently in these compartments (lymph node is enriched in gamma deltaT17, while spleen is enriched in gamma deltaT1), combining these data does not seem warranted. The authors should have provided representative plots for both organs and calculated and analyzed the gamma delta T cell numbers for both organs separately in each of these analyses.*

We did of course process and calculate numbers of different subsets in both lymph nodes and spleen. Where we saw loss of peripheral  $\gamma\delta$  subsets, or rescue, this was reflected in separate analysis of both organs and we did not see any organs specific effects in the mouse strains analysed. We therefore took the initial view that presenting aggregate data was most efficient and least repetitive representation of data. However, we very much recognise the reviewers concern, and interest to see these data, so have now included representative plots across both organs for figure 1D, and show cell numbers of lymph nodes and spleen separately, as well as together, for figures 1, 2, 4 and 7, and these plots reflect the differences observed when we combined data. We did not break down the data for all figures (e.g. figures 3 and 5) as it was more cumbersome for more complex multi-strain comparisons and so attempt to balance clarity and transparency against unnecessary repetitive data presentation.

*(4) The authors make extensive use of surrogate markers in their analysis. While the markers that they choose are widely used, there is a possibility that the expression of some of these markers may be altered in some of their genetic mutants. This could skew their analysis and conclusions. A better approach would have been to employ either*

nuclear stains (Tbx21, RORgammaT) or intracellular cytokine staining to definitively identify functional gamma deltaT1 or gamma deltaT17 subsets.

We did share a similar concern, but think this is not an issue where subsets disappear and are almost completely absent, such as in IKK1/2 KO and Casp8 KO settings. Where we saw rescue with RIPK1<sup>D138N</sup> in Casp8ΔT<sup>CD2</sup> strains, we were keen to demonstrate that the populations we saw restored did exhibit their expected function, and so confirmed this in figure 5C by intracellular cytokine staining after a short 4h restimulation in vitro. This also served to validate our gating strategy, since what we designated as Type 1 cells - CD27+CD122+CD44<sup>int</sup> cells were the only source of IFN-gamma, while CD27-CD44<sup>hi</sup> CD122<sup>lo</sup> cells were the only source of IL-17. Adaptive/ naive cells made neither cytokine. So while we did not include nuclear stains, we were satisfied that the cytokine assays validated the gating strategy.

*(5) The analysis and conclusion of the data in Figure 3A is not convincing. Because the data are graphed on log scale, the magnitude of the rescue by kinase dead RIPK1 appears somewhat overstated. A rough calculation suggests that in type 1 gamma delta T cells, there is ~ 99% decrease in gamma delta T cells in the Cre+WT strain and a ~90% decrease in the Cre+KD+ strain. Similarly, it looks as if the numbers for adaptive gamma delta T cells are a 95% decrease and an 85% decrease, respectively. Comparing these data to the data in Figure 5, which clearly show that kinase dead RIPK1 can completely rescue the Caspase 8 phenotype, the conclusion that gamma delta T cells require IKK activity to repress RIPK1-dependent pathways does not appear to be well-supported. In fact, the data seem more in line with a conclusion that IKK has a significant impact on gamma delta T cell survival in the periphery that cannot be fully explained by invoking Caspase8-dependent apoptosis or necroptosis. Indeed, while the authors seem to ultimately come to this latter conclusion in the Discussion, they clearly state in the Abstract that "IKK repression of RIPK1 is required for survival of peripheral but not thymic gamma delta T cells." Clarification of these conclusions and seeming inconsistencies would greatly strengthen the manuscript. With respect to the actual analysis in Figure 3A, it appears that the authors used a succession of non-parametric t-tests here without any correction. It may be helpful to determine if another analysis, such as ANOVA, may be more appropriate.*

Yes, we completely agree with this assessment and conclusion. While kinase dead RIPK1 does provide some rescue, this appears relatively modest, and instead supports the view, validated in figure 7, that maybe the dominant function of IKK in  $\gamma\delta$  T cells is to activate NF- $\kappa$ B dependent survival signals. Nevertheless, RIPK1<sup>D138N</sup> does provide some significant rescue, which allows some peripheral cells to repopulate and demonstrates that IKK is repressing RIPK1 mediated cell death. It is actually not trivial to assess the relative importance of IKK-RIPK1 and IKK-NF- $\kappa$ B functions. In the IKKΔT<sup>CD2</sup> RIPK1<sup>D138N</sup> mice, we prevent RIPK1 induced death, but still lack the NF- $\kappa$ B-dependent survival signal. Consistent with this, the ~1log reduction in  $\gamma\delta$  numbers between WT and IKKΔT<sup>CD2</sup> RIPK1<sup>D138N</sup> mice is actually similar to what we observe in the absence of REL subunits (Fig. 7) which is a smaller reduction than we observe in IKKΔT<sup>CD2</sup> mice. What would have been ideal is to have a scenario where IKK regulation of RIPK1 was defective but NF- $\kappa$ B survival signalling was intact. This would reveal the full impact of loosening IKK dependent regulation of RIPK1 alone, which we suspect would result in substantial cell death that could not be blocked by NF- $\kappa$ B. Unfortunately, we not have or know of suitable mouse mutants to test this. This is quite a nuanced discussion and we now clarify the scope and extent of conclusions we can draw (p. 7, 11).

*(6) The conclusion that the alternative pathway is redundant for the development and persistence of the major gamma delta T cell subsets is at odds with a previous report demonstrating that Relb is required for gamma delta T17 development (Powolny-Budnicka, I., et al., Immunity 34: 364-374, 2011). This paper also reported the*

*involvement of RelA in gamma delta T17 development. The present manuscript would be greatly improved by the inclusion of a discussion of these results.*

Thank you - we include a discussion of these papers now (p12).

*(7) The data in Figures 1C and 3A are somewhat confusing in that while both are from the lymph nodes of IKKdeltaTCD2 mice, the data appear to be quite different (In Figure 3A, the frequency of gamma delta T cells increases and there is a near complete loss of the CD27+ subset. In Figure 1A, the frequency of gamma delta T cells is drastically decreased, and there is only a slight loss of the CD27+ subset.)*

Yes, we agree these do look quite different and could be confusing. The lymph nodes from IKKΔT<sup>CD2</sup> lack αβ T cells and B cells, and so the cellularity is much lower than normal. Consequently, the percentage representation of remaining cells can be more noisy, while total cellularity calculations are more consistent. This is not an issue in the other strains that all have more cells in lymph nodes. We now show plots from spleen of the same mice which appear better aligned with additional splenic data shown in Figure 1.

**Reviewer #2 (Public review):**

*(1) All approaches used confer changes to the entire T cell compartment. Therefore, the authors are unable to resolve whether the observations are mediated by direct and/or indirect effects (e.g., disorganized lymphoid architecture impacting maintenance/survival/homing).*

We address this important point in the discussion (p11-12). The impacts of gene deletions upon αβ and γδ T cells operate independently of one another (as also discussed in response to reviewer 1). For instance, the phenotype of αβ T cells is identical in IKKΔT<sup>CD2</sup> and IKKΔT<sup>CD4</sup> mice - γδ T cells are only targeted in IKKΔT<sup>CD2</sup> mice. Similarly, the phenotype of γδ T cells is similar in IKKΔT<sup>CD2</sup> vs Casp8.IKKΔT<sup>CD2</sup> strains. αβ T cells are absent from IKKΔT<sup>CD2</sup> but present in near normal numbers in Casp8.IKKΔT<sup>CD2</sup> mice. Others have also noted that γδ T cell development is normal in Rag deficient mice (10.1126/science.1604321). In any case, an absence of αβ T cells is expected to promote γδ T cell survival in the absence of competition for common utilised cytokines such as IL-7 and IL-15, though we do not see much evidence for this in mice with and without αβ T cells such as IKKΔT<sup>CD2</sup> vs Casp8. IKKΔT<sup>CD2</sup> strains. We do now discuss the potential contribution of trans-conditioning for type 17 γδ T cell development (p12).

*(2) Assessment of factors that impact T cell numbers in the periphery is necessary. Are there observable changes to the proliferation, survival, and migration of gd T cell subsets?*

In IKKΔT<sup>CD2</sup> and Casp8. IKKΔT<sup>CD2</sup> deficient strains, we infer a defect in survival, since they lack peripheral γδ T cells, despite normal thymic development. Their absence made it hard to assess proliferation and migration, though γδ T cells were absent from all lymphoid organs. The conclusions that defective survival is responsible for the absence of γδ T cells in the different strains is also supported by the rescue of IKKΔT<sup>CD2</sup> and Casp8ΔT<sup>CD2</sup> strains by kinase dead RIPK1D138N. Furthermore, the presence of small numbers of residual populations in lymph nodes and spleen of IKKΔT<sup>CD2</sup> and Casp8ΔT<sup>CD2</sup> strains demonstrates that migration patterns were normal. Were cells unable to recirculate, they might be expected to fail to leave the thymus, or to accumulate in the spleen. We see no evidence of either of these scenarios.

*(3) TCRd chain usage, especially among type 3 gd T cells, should be assessed.*

We did not unfortunately, assess chain usage, choosing rather to rely of phenotypic identity of specific subsets, which we show in figure 5C, was extremely robust. IL-17 was only secreted



by CD27<sup>−</sup> CD44<sup>hi</sup>  $\gamma\delta$  T cells, while IFN- $\gamma$  was only secreted by CD27<sup>+</sup> CD44<sup>hi</sup>  $\gamma\delta$  T cells. We argue that the production of these key effector cytokines is the most direct test of a subsets functional identity and the phenotypic designation is robust.

*(4) The functional consequences of IKK signaling on gd T cells were largely unaddressed. Cytokine analyses were performed only in the RIPK1D138N Casp8 $\Delta$ TCD2 model, leaving open the question of how canonical NF- $\kappa$ B-dependent signaling impacts the long-term functionality of gd T cells.*

Yes, we agree this remains an open question around the transcriptional mechanisms by which NF $\kappa$ B signalling promotes cell survival, and one best addressed in future studies. We did not perform cytokine staining more widely, because the cytokine assay relies on short term re-stimulation of T cells with PMA and ionomycin. PMA activates PKC which in turn activates NF- $\kappa$ B signalling to elicit the cytokine response measured in this assay. As such, the results of such assays would be hard to interpret. We agree it would be interesting to investigate the functional consequences of REL deficiency in future studies, although this may need a more nuanced setting where  $\gamma\delta$  T cells are not lost as a result of their defective survival.

*(5) The authors suggest that Caspase 8 is required for the development and maintenance of type 3 gd T cells. While the authors discussed the limitations of assessing adult mice in interpreting the data, it seems like a relatively straightforward experiment to perform.*

We did attempt these experiments with collaborators by analysing type 17  $\gamma\delta$  T cell development in fetal thymic organ culture (FTOC). However, the GM mice are not so easy to breed and generating the large numbers of embryos required to set up the FTOCs proved too challenging and we were unable to generate these data.

*(6) While analyses of Casp8 $\Delta$ TCD2 RIPK1D138N mice suggest that loss of adaptive and type 1 gamma delta T cells in Casp8 $\Delta$ TCD2 animals is due to necroptosis, the contribution of RIPK3 kinase activity remains unexamined. RIPK3 activity determines whether cells die via necroptosis or apoptosis in RIPK1/Caspase8-dependent signaling, and inclusion of this analysis would strengthen mechanistic insights.*

Given time and resources, it would have been ideal to confirm necroptotic cell death by alternative knockouts, such as RIPK3 or MLKL. However, formation of the necrosome is dependent on kinase active RIPK1, since autophosphorylation of RIPK1 changes its conformation to allow recruitment of RIPK3 and MLKL and formation of the necrosome. Therefore, the rescue of CASPASE8 deficient T cells from cell death by kinase dead RIPK1 is very solid genetic evidence of necroptosis.

*(7) Canonical NF- $\kappa$ B signaling through cRel alone was not evaluated, leaving a gap in the understanding of transcriptional pathways required for gd T cell subsets.*

This was assessed in p105/RelA knockout strain, which only express cREL. What we lacked was an assessment of what RelA/p50 dimers can support in the absence of cREL. We do however, show the impact of RelA single deficiency, and RelA/p50 deficiency.

In truth, we had many REL deficient strains and it was challenging to make all the combinations we wanted. However, we try to compensate for this by discussing what cREL:cREL dimers and cREL:P50 dimers are capable of doing by analysing  $\gamma\delta$  T cell development in p105/RELA DKO and RELA KO mice - these do show that cREL:P50 can compensate in the absence of RELA, but cREL:cREL cannot.

#### **Reviewer #3 (Public review):**

*Weaknesses:*



*The paper would benefit greatly from a graphical abstract that could summarize the key findings, making the key findings accessible to the general immunology or biochemistry reader. Ideally, this graphic would distinguish the requirements for NF- $\kappa$ B signals sustaining thymic  $\gamma\delta$  T cell differentiation from peripheral maintenance, taking into account the various subsets and signaling pathways required. In addition, the authors should consider adding further literature comparing the requirements for NF- $\kappa$ B /necroptosis pathways in regulating other non-conventional T cell populations, such as iNKT, MAIT, or FOXP3+ Treg cells. These data might help position the requirements described here for  $\gamma\delta$  T cells compared to other subsets, with respect to homeostatic cues and transcriptional states.*

Thank you - we have added such discussions. We are happy to add a graphical abstract if journal constraints permit this.

*Last and least, there are multiple grammatical errors throughout the manuscript, and it would benefit from further editing. Likewise, there are some minor errors in figures (e.g., Figure 3A, add percentage for plot from IKKDT.RIPK1D138N mouse; Figure 7, "Adative").*

Thank you !

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